Remote anesthesia

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Module Objective

- At the end of this module, students will be able to provide day-care anesthesia and procedural sedation techniques for different procedures.

Module Competencies

- Provide procedural sedation for different surgical procedures in a compassionate respectful and caring manner

Learning Outcomes

- Describe basic considerations for office-based anesthesia and procedure outside the operating room.
- Define procedural sedation

- Describe indications for procedural sedation

- Identify what equipment, staffing, and the venue are required before proceeding with the procedural sedation

- Discuss the advantages and risks of procedural sedation
- Perform preoperative evaluation for patients who require procedural sedation
- Apply those procedural sedation guidelines/sedation scales appropriately

- manage complications related to procedural sedation
- Administer sedation agents safely
- Demonstrate how to avoid and respond to adverse events of procedural sedation

- apply the principles of discharging criteria after procedural sedation
- Work in a collaborative way with the other team members
- Apply principles of Enhanced recovery after surgery (ERAS)

Remote anesthesia

Definition

Remote anesthesia is **the provision of anesthesia and sedation outside the theatre environment.**

Center for remote anesthesia

Key points during remote anesthesia

- Standards of anesthesia care and patient monitoring do not vary with the anesthetizing location .
- Open communication between the anesthesiologists /Anesthetists and physicians working in the non operating room anesthetizing location is key to efficient and timely provision of anesthetic care to patients in these areas .

Key points (Miller)

- Anesthesiologists /Anesthetists working in areas were radiology equipment is in use need to be knowledgeable regarding radiation safety and take adequate precautions to ensure their own safety .

- The iodinated contrast media used in radiology and neuro radiology suites as well as the cardiac catheterization laboratory may cause significant adverse reactions and patients receiving contrast media require close monitoring .

Key points

- The unique nature of the magnetic resonance imaging suite with its powerful magnetic fields manifest special care and training to avoid disastrous adverse events .
- Provision of quality anesthesia management in the cardiac catheterization laboratory requires an understanding of the patients underlining conditions and the procedure to be performed.

Key points

- Electroconvulsive therapy has profound physiologic effects that must be understood and dealt with by anesthesiologist /anesthetist providing care to patients undergoing such therapy.

- Provision of anesthetic care in areas where therapeutic radiation is used requires further understanding of radiation safety and is facilitated by the installation of devices to allow remote monitoring of patients under general anesthesia

Introduction

Why remote anesthesia ?

- Recent advances in medical practices such as imaging and minimally invasive diagnostic and therapeutic procedures have led to an exponential growth in the need for anesthesia services outside of the operating room (OR); so-called remote anesthesia.

- There are increasing numbers of surgical diagnostic and therapeutic procedures performed outside of the main theatre environment. These procedures may require anesthetic interventions through monitored care, sedation, regional anesthesia or general anesthesia.

Challenges

- The challenge for anesthesia is to develop a framework that supports and regulates the safe delivery of care.

- Each area presents unique challenges for the anesthesia care team (eg, the patient population served, the nature of the location, the procedures performed).
- The trend of expanding the range of procedures outside of the OR will continue, and the demand for anesthesia services in these locations will grow.

- The complexity and challenges of providing anesthesia care in the non-theatre environment should be acknowledged through appropriate regulation of healthcare providers and training and certification of anesthesia providers. Personnel should be certified resuscitation providers.

- Use of different Techniques such that the development of deep sedation techniques and general anesthesia with total intravenous anesthesia (TIVA)/target-controlled infusion (TCI) techniques(User defined concentration in the central compartment requires computer based calculation) may remove the requirement for complex gas delivery systems and anesthetic machines.

- The anesthesia providers should develop safe practice guidelines that consider the assessment, induction, recovery and discharge of patients. In addition, procedure-specific risks such as radiation exposure and infection control should be considered.
- Compliance with the safe surgery checklist is obligatory. Complication management should be written into patient pathways with consideration of access to other medical, surgical and critical care services.

- The diverse range of locations, procedures and patient groups can provide a challenge to the anesthesia team.

Challenges (Con’t)

- Equipment might be old, not regularly serviced and not in standard use as in the rest of the hospital
- Monitoring standards may not be adequate
- Piped medical gases may not be supplied

- Other personnel may be unaware of the problems facing the anesthetist.

- Space may be limited by bully equipment making access to the patient difficult

Challenges related to the environment

- Poor environmental conditions (eg Lighting, temperature)
- Recovery facilities may not be available
- Inadequate ventilation/scavenging causing pollution
- Problem related to transferring patients

Challenges related to patient

Patients who require general anesthesia are

- Infants or uncooperative children
- Older children or adults with psychological behavioral or movement disorders
- Intubated patients such as acute trauma victims and patients receiving intensive care
- Interventional procedures radio-guidance or painful procedures like ECT, cardioversion which require amnesia

Challenges related to the procedure

- MRI related problems
- Bleeding
- Conversion from sedation to anesthesia
- Contrast related problems
- Radiation

Way forward

Anesthetists must maintain the same high standards as in the operating room, which requires appropriate facilities and staff, as well as suitable pre and post anesthesia care.

They must ensure that the same standards and codes that govern the OR and the recovery of patients from anesthesia are present in these remote locations before administration of an anesthetic.

Equipment for the Remote Location

Primary and backup oxygen supply (checked O_2 tank), and means to provide intermittent positive-pressure ventilation (IPPV).

- Facilities for gas scavenging if any inhalation anesthetics are used.
- Functioning suction apparatus.

Equipment's cont.'s

- Adequate lighting.
- Electrical outlets (operating room standards).
- Means of immediate communication to the operating room personnel.
- Facilities and staff for the preparation and recovery of children/Adults according to published guidelines.

Equipment's cont.'s

- Resuscitation cart and defibrillator (immediately available).
- In addition, all drugs and equipment to manage the child during the procedure must be provided:
- Monitoring for pulse oximetry, electrocardiography (ECG), end-tidal carbon dioxide, blood pressure, and body temperature.
- Airway supplies; age and size appropriate facemasks, laryngoscope blades, endotracheal tubes, oropharyngeal airways, laryngeal mask airways, suction catheters, and breathing circuits.

- Appropriate anesthesia equipment, infusion pumps, etc.
- All necessary drugs for anesthesia and sedation, and, syringes, a variety of IV catheters, intravenous fluid and fluid administration sets.
- Emergency resuscitation drugs.
- Where indicated: equipment to maintain body temperature (e.g., forced air warmer for cardiac catheterization laboratory).

EQUIPMENTS CHECK: “SOAPME”

- S (suction) – Appropriate size suction catheters and

functioning suction apparatus.

- O (oxygen) – Reliable oxygen sources with a functioning flowmeter.

- A (airway) – Size appropriate airway equipment's

- P (pharmacy) –Basic drugs needed for life support
during emergency
- M (monitors)

- E (Equipment's)

General Principles

- The technique chosen should result in minimal (if any) post anesthesia squeal.
- Use short-acting drugs that will not delay recovery and are not associated with postoperative nausea and vomiting.

General Principles(Con ’t)

- Most children can be managed with an oxygen mask or nasal prongs, and a support to extend the neck. Occasionally an oral or nasopharyngeal airway may be required.
- The laryngeal mask airway (LMA) may be a useful alternative. Finally, if all of the above maneuvers still result in an obstructed airway, tracheal intubation may be required.

In general, tracheal intubation is avoided in remote locations to minimize post extubation airway problems and to abbreviate the recovery time.

General Principles(Continued)

- When repeated anesthetics will be needed, a chronic intravenous line should be maintained: either a central line or a “Hep-Locked” peripheral line.
- Monitor every child or Adult as you would in the operating room.

- patients evaluation before the procedure by qualified personnel, coexisting medical conditions, fasting status, informed consent.

- Standard monitoring practice: level of consciousness, ventilation, oxygenation, and hemodynamics .

- Designated personnel: for monitoring and maintenance of sedation and analgesia

- BLS trained personnel should be present
- Supplemental oxygen ,emergency equipment's, drugs, defibrillator
- Adequate recovery care should be provided.

Monitoring

- Oxygenation, ventilation, circulation, and temperature should be continually evaluated.

- Oxygen concentrations of inspired gas: low-concentration alarm
- Blood oxygenation : pulse oximetry.
- Ventilation : observation of the patient and end-tidal carbon dioxide.

- Circulation : ECG, arterial blood pressure (5 minutes)
- Temperature
- ASA GUIDELINES

- Adequate monitoring equipment to meet the standards for basic anesthetic monitoring and A self inflating hand resuscitator bag .

- Sufficient safe electrical outlets
- Adequate patient and anesthesia machine illumination with battery powered backup
- Sufficient space for the anesthesia care team

Emergency cart with

- defibrillator, Emergency drugs, and other emergency equipment

- A means of reliable two-way communication to request assistance

- Compliance of the facility with all applicable safety and building codes .

Anesthesia techniques in remote locations:

- Minimal, moderate and deep sedation/ analgesia
- General anesthesia
- Minimal sedation/analgesia: level of anesthesia at which <u>patients are able to tolerate unpleasant procedures through relief of anxiety, discomfort, or pain.</u>

- Moderate sedation/analgesia: <u>patient retains purposeful responses to stimulation, requires no airway intervention, and can maintain adequate ventilation and cardiovascular function</u>.

- <u>Deeper levels of sedation/General anesthesia</u>: requires the presence of adequately trained anesthesia personnel

Medications

- Combinations of sedative and analgesic medications, such as benzodiazepines and opioids, provide effective moderate sedation
- Doses should be carefully titrated
- Intravenous access maintained throughout the procedure.

- When oral agents are used (e.g., benzodiazepines, chloral hydrate), adequate time must be given to allow complete drug uptake before supplementation can be considered.
- Propofol, ketamine, thiopentone can also be used
- Antagonists should be available: Naloxone and flumazenil

- Wherever anesthesia or sedation is undertaken, a full range of emergency drugs including specific reversal agents such as naloxone, sugammadex(used to reverse the effect of muscle relaxants) and flumazenil should be made available.

- In remote locations where anesthesia is undertaken, drugs to treat rare situations, such as dantrolene for malignant hyperthermia, or intralipid for local anesthetic toxicity should be immediately available and located in a designated area.

- Robust systems should be in place to ensure reliable medicines management, including storage facilities, stock review, supply, expiry checks, and access to appropriately trained pharmacy staff to manage any drug shortages

- All local anesthetic solutions should be stored separately from intravenous infusion solutions, to reduce the risk of accidental intravenous administration of such drugs.

Recovery care

- The patient must be medically stable before transport.
- Must be accompanied to the recovery area by the individual providing the anesthesia or sedation/analgesia care.
- Provision of oxygen delivery and monitoring while the patient is on the transport cart may be required.
- Appropriate recovery facilities and staff must be provided

- In the recovery area, the patient's condition must be documented and continually assessed. (Use of Alde ret score)respiration ,circulation ,consciousness ,color and level of activity (Like APGAR 1970 modified score)
- Immediate availability of personnel trained in advanced cardiac life support should be ensured.

Procedures
Burn Wound dressing Change

- Definition –Burn is type of injury to the tissue Burns can be thermal, chemical, radiation and electrical.

- Burns are a serious public health problem throughout the world. In 2002 fire-related burns were responsible for an estimated 322,000 deaths worldwide.

- Deaths are only part of the problem; for every person who dies, many more are left with lifelong disabilities. Over 95% of fatal fire-related burns occur in low and middle income countries.
- The standard of burn care that is routinely available in many countries, is suboptimal due to lack of education and resources

Risk Factors

- Open fires for cooking, heating and lighting.
- Substance abuse including alcohol and smoking.
- Low socioeconomic status, overcrowding, lack of safety measures, lack of parental supervision.
- Medical conditions such as epilepsy.

- Burns are a major cause of morbidity and mortality. .

- Initial care should follow the basic principles of trauma resuscitation, with assessment and treatment of life threatening problems of the airway, breathing and circulation. Recognition of burn severity, stopping the burning process, initiating fluid and analgesia and early surgical involvement can have a significant effect on patient outcome

- The usual cause of death in burned patients is Burn Shock and Infection
- During severe burn injury hemodynamics in the early phase are characterized by a reduction in cardiac output and increased systemic and pulmonary vascular resistance with or without pulmonary edema.
- Approximately 3 to 5 days after major burn injury, a hyper dynamic and hyper metabolic state begins with tachycardia, increased stroke volume, hyperthermia, and increased protein catabolism.

Challenges in burned patients

- Sepsis systemic inflammatory response syndrome altered pharmacokinetic compartment and receptor population changes makes drugs use unpredictable

- Hyperkalemia following succinylcholine

- Resistance to Non depolarizing muscle relaxants

- Airway compromise leads to difficult airway management
- Vascular access. prolonged procedures ,Blood loss and Temperature hemostasis are major challenges .

Burned patients

- All burn patients should be approached initially as multiple-trauma patients.
- Inhalation injury is a major source of mortality in burn patients. If the history and physical examination suggest inhalation injury, one should have a low threshold for early intubation.

Resuscitation in burned patients

- Multiple fluid resuscitation formulas exist for estimating fluid needs and differ somewhat in their recommendations for the amount of crystalloid and colloid replacement

- The Parkland formula, one of the most popular resuscitation regimens, recommends using 4 mL per percentage total body surface area (%TBSA) burn per kilogram administered over the first 24 hours, with half of this calculated volume administered during the first 8 post injury hours. The remaining half is administered over the next 16 hours.

- 30%burn surface area calculations -------------

- The magnitude of burns is classified according to percentage of total body surface area (TBSA) involved, depth of the burn, and the presence or absence of inhalational injury.
- TBSA burned in adults can be estimated using the "rule of 9s," an age-specific diagram, or by estimation using the palmar surface of the hand.

- Major burn injury results in pathophysiologic changes in virtually all organ systems.

- The perioperative care of burn patients requires knowledge of these changes from the initial period of injury through the period when wounds are covered and healed.

- In patients with severe burn injury, massive heat loss can occur through the open wounds, making the maintenance of Normothermia challenging. Multiple strategies are used to maintain body temperature in the operating room, including high ambient room temperatures, use of warming blankets, radiant warmers, blood/fluid warmers, and wrapping the head and extremities with plastic or thermal insulation.

- Securing the endotracheal tube in a patient with facial burns can be difficult. It is essential to secure the endotracheal tube with a carefully secured harness to avoid the potential catastrophe of accidental extubation.

- Burn patients develop tolerance to most narcotics and sedatives, thereby requiring higher doses than patients without thermal injury. Sedatives and narcotics should be titrated to effect while the patient is carefully monitored

- In burn patients, exposure to succinylcholine can result in an exaggerated hyperkalemic response that can induce cardiac arrest.

- The general recommendation is to avoid succinylcholine administration in patients beginning 24 to 72 hours after burn injury.

- The duration of this dangerous response to succinylcholine after burn injury is unknown

- Blood loss during burn wound excision can be deceptively large. It is not difficult for the surgical team to remove eschar so rapidly that the patient becomes hypovolemic and unstable.

- Good communication between the surgical and anesthesia teams as well as limiting the operative duration and extent of excision can prevent such problems.

- Burn injury leads to increased susceptibility to infection through multiple mechanisms including loss of the physical barrier of an intact skin, damage to lining of the respiratory tract from inhalation, and altered gut permeability and function.
- Maintaining strict aseptic techniques both in the operating room and during transport is essential.

- Nearly every aspect in the treatment of burn injury is associated with pain. Poorly controlled pain and anxiety can have significant adverse physiologic and psychological effects.

- Standardized pain and anxiety guidelines are used to provide appropriate, consistent patient comfort.

- Standard care of burn wounds consists of cleaning and debridement (removing devitalized tissue), followed by daily dressing changes

- The adverse squeal of inadequate pain control in the burn population have been long recognized, yet control of pain remains inadequate globally.

- The dynamic evolution of burn pain both centrally and peripherally, and the many factors which influence pain perception illustrate the need for a therapeutic plan which is similarly dynamic and flexible enough to cope with the facets of background, breakthrough, procedural and post-operative pain

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- Children and Adults with burns undergo multiple, painful and anxiety-provoking procedures during wound care and rehabilitation;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;

- The goal of procedural sedation is safe and efficacious management of pain and emotional distress, requiring a careful and systematic approach.

- Achieving the best results needs understanding of the mechanisms of pain and the physiologic changes in burn patients, frequent evaluation and assessment of pain and anxiety, and administration of suitable pharmacological and non pharmacological therapies. Pharmacological therapies provide the backbone of analgesia and sedation for procedural pain management.

- Opioids provide excellent pain control, but they must be administered judiciously due to their side effects. Sedative drugs, such as benzodiazepines and propofol, provide excellent sedation, but they must not be used as a substitute for analgesic drugs.

- Ketamine is increasingly used for analgesia and sedation in children as a single agent or an adjuvant.
- Non pharmacological therapies such as virtual reality(Simulators are used), relaxation, cartoon viewing, music, massage and hypnosis are necessary components of procedural sedation and analgesia for children.

- Pain and sedation management for patients undergoing burn dressing change can be challenging.
- Variations appear to exist in the selection of medications before and during burn dressing change

- In outpatient settings, benzodiazepines, ketamine, oral opioids, and topical lidocaine were used frequently, as were non pharmacologic methods of distraction and comfort. Sedation in pediatric burn patients is challenging.
- Responses highlighted areas for improvement regarding pain control during dressing changes and increasing use of multimodal analgesia.

- Commonly used medications including opioids, benzodiazepine, and ketamine are well established in the treatment of burn patients, as are non pharmacologic methods.

- A collaborative effort among institutions is needed to formulate practice guidelines for sedation during burn dressing.

Computed Tomography

- It is commonly used for diagnostic and invasive therapeutic
- Contrast media is used to enhance the quality of image.

Radiation safety:

- \- Dosimeters should be worn
- \- Lead apron and thyroid shields
- \- Movable leaded glass screens
- \- Video monitoring

- Remote mirroring of monitor data

- In the United States, the maximal permissible radiation dose for occupationally exposed persons is 50 millisieverts (mSv) annually,

- lifetime cumulative dose of 10 mSv × age, and monthly exposure a 0.5 mSv for pregnant women

Reactions to iodinated contrast media

Predisposing factors :

History of bronchospasm, history of allergy, underlying cardiac disease, hypovolemia, hematologic disease, renal dysfunction, extremes of age, anxiety, and medications such as β-blockers, aspirin, and nonsteroidal anti-inflammatory drugs.

Treatment is symptomatic

Corticosteroids and antihistamines are given to symptomatic patients under the assumption that the cause is immunologic.

Patients who have had previous reactions to contrast media may benefit from pretreatment with prednisolone, 50 mg 12 and 2 hours before a procedure requiring contrast media, and diphenhydramine, 50 mg immediately before the procedure.

Reactions to contrast Iodinated media

Mild		Severe		Life threatening
Nausea Retching perception of warmth		Rigorous feel faint Chest pain		Glotic edema Pulmonary edema
		Vomiting		Life threatening arrhythmia
Head ache		Bronchospasm		Cardiac arrest
Mild Urticarial Itchy rash		Dyspnea Severe Urticarial		Seizure ,unconsciousness
		Abdominal pain arrhythmias Diarrhea Renal failure		

Anesthetic concerns for Computed Tomography

1. Limited patient access and visibility
2. Absolute need to exclude ferromagnetic components
3. Interference/malfunction of monitoring equipment
4. Potential degradation of the imaging caused by the stray RF currents from monitoring equipment and leads

RF (Radio Frequency) current is high-frequency alternating current (AC) that oscillates rapidly, allowing it to generate and propagate electromagnetic waves

5. The necessity to not move the anesthetic and monitoring equipment when the examination has started to prevent degradation of magnetic field homogeneity

6. Limited access to the MRI suite for emergency personnel in accordance with recommended policies .

Anesthetic monitoring concerns

- 1 Electrocardiography: Voltage induced by blood flow in the aorta when the patient is in a static magnetic field produces T and ST-wave abnormalities
- 2. Pulse oximetry: The "antenna effect" in conventional pulse oximetry wires results in thermal injury.
- 3. Noninvasive blood pressure monitoring: connections of the blood pressure cuff and hoses should be plastic.

- 4. Precordial and esophageal stethoscopes: interference by the noise of the scanner during operation.

- 5. Capnography: length of the sample tube: delay in signal transduction.
- 6. Temperature: temperature probes with RF filters to be used(Radio frequency (RF) is a measurement representing the oscillation rate of electromagnetic radiation spectrum, or electromagnetic radio waves, from frequencies ranging from 300 gigahertz (GHz) to as low as 9 kilohertz (kHz)).

ANESTHESIA FOR INTERVENTIONAL NEURORADIOLOGY

Commonly performed procedures:

- Embolization of cerebral and dural AVM's (**arteriovenous malformation**,)
- Coiling of cerebral aneurysms
- Angioplasty of atherosclerotic lesions
- Thrombolysis of acute thromboembolic stroke

General considerations

Procedures requires:

- Deliberate hypotension /hypertension
- Deliberate hypercapnia
- Deliberate cerebral ischemia
- Rapid transition between deep sedation and an awake responsive state

- • High resolution fluoroscopy + DSA- real time images: high exposure of radiation
- Contrast media used

Anesthetic management Airway examination must: intra op. manipulation not possible

- History of contrast media reaction
- Evaluation of hypertension: deliberate hypo/hypertension

required

- • Patient completely immobile during procedure YET awake for neurologic testing
- • Urinary catheters: contrast /osmotic diuretic used frequently
- • Coagulation profile check: heparin commonly used (

Anesthetic management

- Airway examination must: intra op. manipulation not possible
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- Urinary catheters: contrast /osmotic diuretic used frequently

- Coagulation profile check: heparin commonly used (ACT 2-2.5 baseline)

Smooth emergence essential: avoid coughing/bucking

Anesthetic management

- General anesthesia and conscious sedation are both suitable techniques for interventional neuroradiology depending on the complexity of the procedure,

- The need for blood pressure manipulation, the need for intraprocedural assessment of neurologic function

- Propofol infusions or combinations of a benzodiazepine and opioid are frequently used.

Interventional cardiology

- Coronary angiography
- Percutaneous cardiac interventions
- - Cardiac catheterization
- - Coronary artery angioplasty/ stenting
- - Valvotomy

- Endovascular closure of intracardiac defects, cardiac valve replacement
- Electrophysiological studies with pathway ablation
- Cardioversion
- Coronary angiograph

Coronary angiography

- Catheter has been inserted through the femoral artery, brachial, radial, or ulnar arteries.
- Supplemental oxygen is administered, typically by nasal cannula,
- Standard ASA monitors are used.

- Arterial blood pressure can be directly transduced from the arterial introducer
- The usual anesthesia management is by sedation/analgesia,with general anesthesia reserved for failure of sedation
- Anesthetic agents used commonly include fentanyl and midazolam, sometimes supplemented with propofol or dexmedetomidine.

- Heparin is frequently administered, even for diagnostic catheterization, and the effects are often reversed with protamine.
- Typical heparin doses range from 2500 to 5000 IU intravenously.
- For interventional procedures, higher heparin doses (i.e., 10,000 IU intravenously) are given, with a target ACT of greater than 300 seconds.

- After protamine administration :anaphylactic and anaphylactoid reactions

- Platelet aggregation inhibitors and LMWH Provocative agents: Ergon ovine maleate, methylergonovine maleate, or intracoronary acetylcholine : may be given to provoke spasm while contrast medium is injected into the coronary artery.

- Intracoronary vasodilators such as nitroglycerin may be required when provocative agents are administered.

- Intracoronary diltiazem may be used to treat resistant vasospasm

PTCA (Percutaneous trans luminal coronary angioplasty)

- During ischemia and frequently during reperfusion after dilation of the stenotic coronary artery, ventricular arrhythmias may develop and require treatment.

- Hemodynamically significant PVC s and non sustained VT should be initially treated with amiodarone/ cardioversion

- Rupture of the coronary artery may result in hem pericardium and pericardial tamponed : emergency pericardiocentesis,

- Vascular spasm may often be relieved by the injection of 200 µg of nitroglycerin through the coronary artery.
- When a thrombus has formed, intracoronary injection of thrombolytic agents, such as urokinase, may dissolve the thrombus.

PTCA (Percutaneous trans luminal coronary angioplasty

- These patients may require relatively massive autologous platelet transfusions should emergency cardiac surgery be necessary
- Acute coronary occlusion may not respond to transluminal treatment in the catheterization laboratory and may require emergency coronary artery bypass grafting (CABG).
- The patient may have angina, hypotension, and arrhythmias and may benefit from the insertion of an intra-aortic balloon pump.

- Anesthesiologists must be aware of the current protocols for pharmacologic regulation of hemostasis and intimal proliferation for the various procedures performed at their institution because they are evolving constantly.
- Developments in perfusion technology have increased the portability of temporary ventricular assist devices to maintain hemodynamic stability.

Trans esophageal echocardiography is also used for ventricular monitoring, as well as cannula placement.

Pediatric Cardiac Catheterization

General concerns:

- Cardiac anomalies vary from relatively simple atrial septal defects to complex congenital cardiac anomalies
- Shunts may be present at multiple levels, and patients may be profoundly cyanotic
- Patients may also have coexisting non cardiac congenital abnormalities young patients may be uncooperative

- “Steady state” must be maintained for diagnostic accuracy, Even in cyanotic patients, supplemental oxygen is not administered unless oxygen saturation falls below baseline levels .

- – Care must be taken to maintain ventilation and $PaCO_2$ within normal physiologic limits to avoid alterations in PVR

- – Medications : fentanyl, midazolam, propofol, and ketamine

- Premedication with midazolam, 0.5 mg/kg orally
- – Infants and small children : GA preferred
- – If intravenous access is not present: induction with inhaled nitrous oxide, oxygen, and a volatile anesthetic such as sevoflurane is performed.

- Anesthesia is maintained with volatile anesthetics and controlled ventilation with room air
- Controlled ventilation avoids the increases in $PaCO_2$
- Close monitoring
- Repeated ABGs

- - Hypocalcemia and hypoglycemia
- - Hypothermia (Rectal temperature monitoring)
- - Hematocrit monitoring
- • Chronic cyanotic(Polycythemia) ; sufficient fluid should be given to balance osmotic effects of contrast media{ avoid micro embolic events and hem concentration}

Complications of cardiac catheterization

- Bleeding at vascular access sites,
- Perforation of cardiac chambers or great vessels by catheters
- Vascular dissection or hematoma

- Second- or third-degree heart block
- Embolic phenomena
- Arrhythmias
- Sinus bradycardia
- Pericardial tapenade

Pacemaker and Cardioverter-Defibrillator Implantation

- GA is necessary during testing of the ICD
- When the ICD is tested, ventricular tachycardia or fibrillation is induced, and the ability of the device to sense the arrhythmia and deliver appropriate cardioversion-defibrillation energy is confirmed.
- Return of hemodynamic variables to baseline after cardioversion defibrillation must be closely monitored: poor ventricular functions

Complications (during removal of infected /malfunctioning ICD)

- – Rupture of the superior vena cava, right atrium, or right ventricle
- –Severe TR may result from removal of right ventricular leads adherent to the valvular or subvalvular apparatus
- – Leads in place for longer than 1 year may have extensive fibrosis, which may make removal difficult.
- Vascular occlusion

Elective cardioversion

- Elective cardioversion is uncomfortable, and general anesthesia is required
- The purpose of cardioversion is to deliver a precisely timed electrical current to the heart to convert an organized rhythm to a more hemodynamically stable rhythm.(Atrial fibrillation /AF
- The purpose of defibrillation is to deliver a randomly timed high-energy electrical current to the heart to restore a normal sinus rhythm.(VF/Vtachycardia

- In the case of chronic atrial fibrillation, echocardiography is performed before cardioversion to rule out the presence of left atrial thrombi, which could cause stroke
- Standard anesthesia technique and monitoring is indicated

Electroconvulsive therapy (ECT)

Indications

- – Major depression
- – Mania
- – Certain forms of schizophrenia
- – Parkinson’s syndrome

Contraindications

- Pheochromocytoma
- – Increased ICP
- – Recent CVA

- \- Cardiovascular conduction defects
- \- High risk pregnancy
- \- Aortic and cerebral aneurysms

Electroconvulsive therapy (ECT)

Principle:

- ECT consists of programmed electrical stimulation of the central nervous system to initiate seizure activity.
- The seizure is monitored by observation of the patient and by an electroencephalogram on the ECT machine.
- Periods: 6 to 12 treatments over 2 to 4 weeks

physiologic effects:

- Increases in cerebral blood flow and intracranial pressure.
- Initial parasympathetic discharge manifested by bradycardia, occasional asystole, premature atrial and ventricular contractions.
- Hypotension and salivation may be noted. Followed by sympathetic discharge associated with tachycardia, hypertension, PVCs, and rarely, ventricular tachycardia.
- ECG changes, including ST-segment depression and T-wave inversion

General considerations of ECT

- Antidepressants : TCA (block reuptake of catecholamines)- anticholinergic

- Antihistaminic, sedative, slow cardiac conduction

- MAOI (blocks metabolism of catecholamines)- hypertensive crises, inhibit hepatic microsomal enzymes .

- Lithium-prolongs NMB , bzd, barbiturates duration;cognitive effects post ECT (discontinue pre ECT)
- Newer antidepressents (trazadone, bupripion, fluxetine): less side effects; preferred

Anesthetic Management of ECT

- Pre operative evaluation: CNS/ CVS evaluation, Osteoporosis

Anesthesia for ECT

- Anticholinergic pretreatment-glycopyrrolate/atropine:To prevent transient asystole, bradycardia, antisialogogue
- Induction : Intravenous anaesthetics Methohexital (.75-1 mg/kg)---Gold standard   Propofol,etomidate, thiopental, Bzd
- ketamine not usually a choice

Volatile anesthetics-Sevoflurane agent of choice in children specially

- Opoids -Remifentanil
- Neuromuscular blockers: Sch 0.5 mg/kg, (DOC)

Mivacurium :alternative to Sch, but not be as effective as Sch in preventing tonic-clonic muscle contractions ;

Atracurium/ cisatracurium:pseudocholinesterase deficiency.

Esmolol and labetalol have been successfully used to control hypertension and tachycardia after ECT: routinely not recommended because the hypertension and tachycardia are usually self-limited.

Therapeutic Radiation

Intraoperative Radiation Therapy

- IORT is the delivery of radiation during exposure of a tumor or tumor bed during an operative procedure

- Brachytherapy with temporary or permanent implantation of radioactive seeds .

Intraoperative radiation

- IORT can be effectively used as a supplemental boost to external beam treatments

Advantage: increase the tumor dose with less damage to adjacent healthy tissues and more accurate localization of the radiation field.

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- Disadvantages :an operation is needed, optimal dose combinations of external beam radiation and IORT are complex to calculate

- Complications : pain, nausea and vomiting, bowel dysfunction, ureteral obstruction, neuropathy, abscesses, and delayed wound healing

Therapeutic Radiation

Tumors treated with IORT have included locally advanced colorectal cancer, retroperitoneal sarcomas, limb sarcomas, gynecologic malignancies, and pediatric malignancies

- GA with endotracheal intubation required.

GI PROCEDURES

- ENDOSCOPY-• - endoscope is passed into the GI tract

  -evaluates mucosa of the esophagus, stomach & duodenum -dilation of stricture areas.

- COLONOSCOPY -scope is inserted into the rectum evaluate the colon

- ERCP(Endoscopic retrograde cholangiopancreatography)
- Diagnosis of obstructive, neoplastic, or inflammatory pancreatobillary structures

Anesthesia for GI Procedures

- • Pre anesthetic assessment

Type of anesthesia:

- – Moderate sedation- Midazolam and Fentanyl
- – Deep sedation- Addition of propofol Some cases required general anesthesia

Anesthetic considerations:

- – Strong vagal nerve stimulation as result of stimulation to colon Most patients tolerate these procedures well(Vesovagal reflex);;;;;;;;;;;;;;;;;;;;;

Dental Procedures

- • Pediatric Dentistry- fillings, crowns, pulpotomies, tooth extractions and space maintainers
- Oral and Maxillofacial Surgery- extractions of impacted teeth, insertion of dental implants, treatment of infections of the head and neck and facial cosmetics
- Periodontics- surgery of teeth, gingiva, connective tissue,
- periodontal ligament and alveolar bone
- Anesthesia : general anesthesia, minimal sedation, moderate sedation with local anesthetic for particular areas of surgey

Urological procedures

- Extracorporeal Shock Wave Lithotripsy-- kidney and ureteral stones
- Cystoscopy/ ureteroscopy-- diagnosis and treat lesions of the lower (urethra, prostate, bladder) and upper (ureter, kidney) urinary tracts.

Anesthesia Technique

Depending on the patient and procedure anesthesia

- • Topical lubrication
- • MAC
- • Regional

Regional Anesthesia

- • T-6 level - upper tract instrumentation
- • T-10 level – lower tract surgery

Clinical pearls

- Standards of anesthesia care and patient monitoring do not vary with the anesthetizing location.
- Open communication between anesthesiologists/Anesthetists and physicians working.
- Anesthesiologists/Anesthetist should be knowledgeable regarding radiation safety and take adequate precautions to ensure their own safety.

- The iodinated contrast media used in the radiology and neuroradiology suites, as well as the cardiac catheterization laboratory, may cause significant adverse reactions

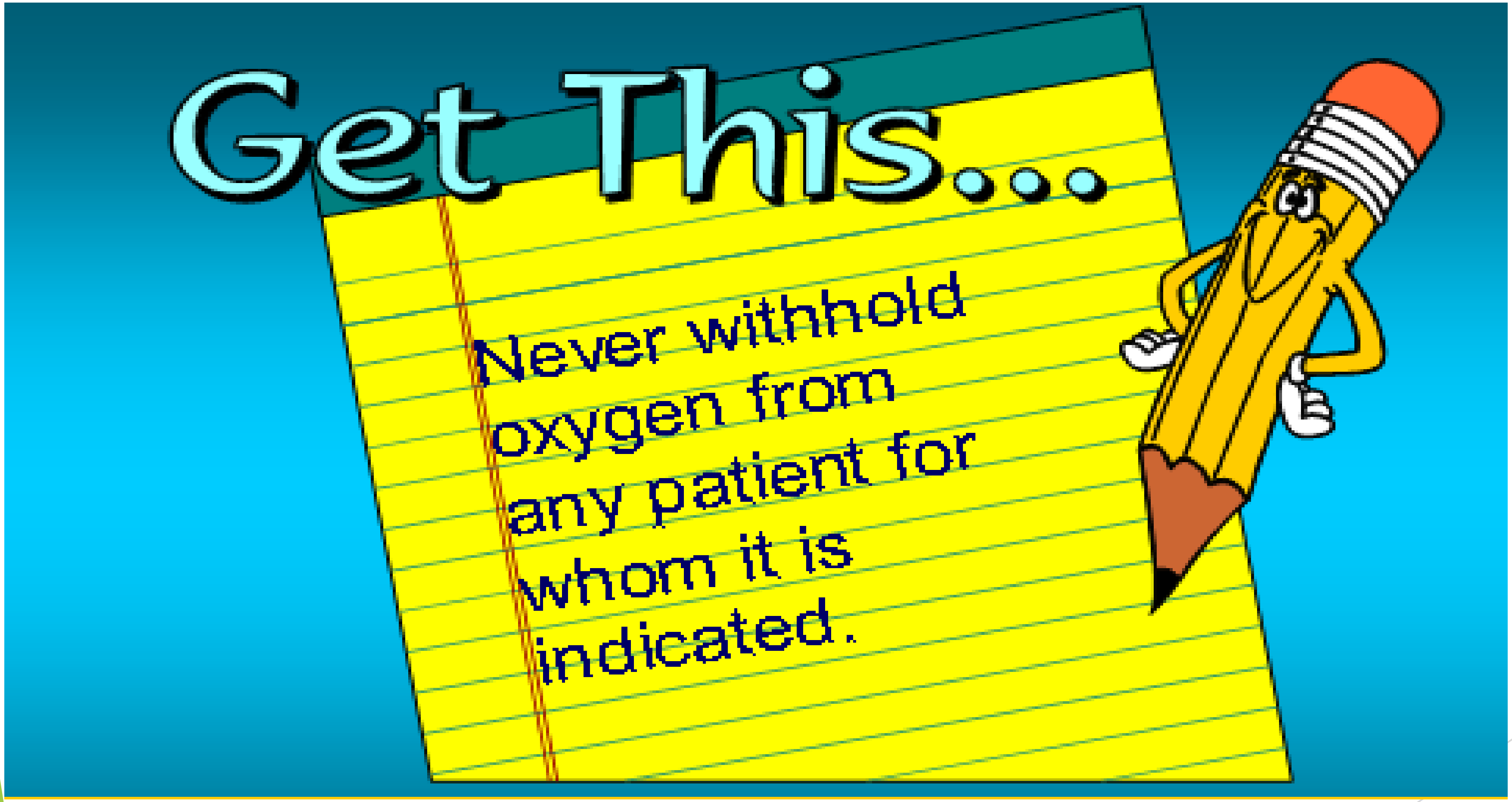
- The unique nature of the magnetic resonance imaging suite with its powerful magnetic fields mandates special care

- Provision of quality anesthetic management in the cardiac catheterization laboratory requires an understanding of the patient's underlying condition and the procedure to be performed.

Electroconvulsive therapy has profound physiologic effects that must be understood and dealt with by anesthesiologists /Anesthetists providing care to patients undergoing such therapy.

Provision of anesthesia services in areas where therapeutic radiation is used requires further understanding of radiation to allow remote monitoring of patients under general anesthesia.

- Preoperative and postoperative management of pulmonary problems in elderly patients is of particular importance in the prevention of morbidity and mortality in Remote locations .


Get This...
Never withhold oxygen from any patient for whom it is indicated.

THANK YOU